Solution to the Daily Limit

Based on the problem provided in the file dailyintegral-limit-limit-1x1-2026-08-21.jpg, we are tasked with finding the limit:

$$\lim_{n\to\infty} n^2(A - A_n)$$

where $\mathcal{E}$ is the ellipse defined by $\frac{x^2}{9} + \frac{y^2}{4} = 1$, A is the area of $\mathcal{E}$, and A_n is the maximum possible area of an n-sided polygon inscribed in $\mathcal{E}$.

1. Area of the Ellipse and the Inscribed Polygon

From the equation of the ellipse $\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$, we can identify the semi-major axis $a = 3$ and the semi-minor axis $b = 2$. The area of the ellipse, A, is given by the standard formula:

$$A = \pi ab = \pi(3)(2) = 6\pi$$

To find A_n, we can use the property that an ellipse is an affine transformation of a unit circle (specifically, scaled by a in the x-direction and b in the y-direction). Because affine transformations preserve the ratios of areas, the maximum area of an n-sided polygon inscribed in the ellipse corresponds to the transformed maximum area n-sided polygon inscribed in a unit circle.

The maximum area of an n-sided polygon inscribed in a unit circle is achieved by a regular n-gon. The area of a regular n-gon inscribed in a unit circle is composed of n isosceles triangles with side lengths 1 and an included angle of $\frac{2\pi}{n}$, giving:

$$\text{Area of regular } n\text{-gon} = n\left(\frac{1}{2}(1)(1)\sin\left(\frac{2\pi}{n}\right)\right) = \frac{n}{2}\sin\left(\frac{2\pi}{n}\right)$$

Applying the scaling factor ab to get the maximum area inscribed in the ellipse, we have:

$$A_n = ab\left[\frac{n}{2}\sin\left(\frac{2\pi}{n}\right)\right] = (3)(2)\frac{n}{2}\sin\left(\frac{2\pi}{n}\right) = 3n\sin\left(\frac{2\pi}{n}\right)$$

2. Taylor Series Expansion

We are taking the limit as $n \to \infty$, which means the argument of the sine function, $\frac{2\pi}{n}$, approaches 0. We can use the Maclaurin series expansion for $\sin(x)$:

$$\sin(x) = x - \frac{x^3}{3!} + \mathcal{O}(x^5)$$

Substituting $x = \frac{2\pi}{n}$:

$$\begin{aligned}
\sin\left(\frac{2\pi}{n}\right) &= \left(\frac{2\pi}{n}\right) - \frac{1}{6}\left(\frac{2\pi}{n}\right)^3 + \mathcal{O}\left(\frac{1}{n^5}\right) \\
&= \frac{2\pi}{n} - \frac{8\pi^3}{6n^3} + \mathcal{O}\left(\frac{1}{n^5}\right) \\
&= \frac{2\pi}{n} - \frac{4\pi^3}{3n^3} + \mathcal{O}\left(\frac{1}{n^5}\right)
\end{aligned}$$

Now, substitute this expansion back into the expression for A_n:

$$\begin{aligned} A_n &= 3n\left(\frac{2\pi}{n} - \frac{4\pi^3}{3n^3} + \mathcal{O}\left(\frac{1}{n^5}\right)\right) \\ &= 6\pi - \frac{4\pi^3}{n^2} + \mathcal{O}\left(\frac{1}{n^4}\right) \end{aligned}$$

3. Evaluating the Limit

Now we evaluate the required expression $A - A_n$:

$$\begin{aligned} A - A_n &= 6\pi - \left(6\pi - \frac{4\pi^3}{n^2} + \mathcal{O}\left(\frac{1}{n^4}\right)\right) \\ &= \frac{4\pi^3}{n^2} - \mathcal{O}\left(\frac{1}{n^4}\right) \end{aligned}$$

Multiply by n^2 as required by the limit:

$$\begin{aligned} n^2(A - A_n) &= n^2\left(\frac{4\pi^3}{n^2} - \mathcal{O}\left(\frac{1}{n^4}\right)\right) \\ &= 4\pi^3 - \mathcal{O}\left(\frac{1}{n^2}\right) \end{aligned}$$

Taking the limit as $n \to \infty$:

$$\lim_{n\to\infty} n^2(A - A_n) = \lim_{n\to\infty}\left(4\pi^3 - \mathcal{O}\left(\frac{1}{n^2}\right)\right) = 4\pi^3$$

Conclusion

Therefore, the evaluated limit is:

$$\lim_{n\to\infty} n^2(A - A_n) = 4\pi^3$$